Attention, Perception, & Psychophysics

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Journal:	*Attention, Perception, & Psychophysics*
Manuscript ID	Draft
Manuscript Type:	Original Manuscript
Date Submitted by the Author:	n/a
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Keywords:	speech perception, spoken word recognition, evoked potentials

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Words Disappear from Perceptual Processing

Abstract

Distal speech rate, the rate of speech context which is nonadjacent to a to-be-detected word, has a large effect on whether listeners report hearing acoustically blended words lacking clear amplitude or spectrotemporal cues to word onset. Event-related potentials (ERP) were employed to determine if distal speech rate affects early perceptual processing of reduced function words or only later interpretation of processed signals. Consistent with previous results, slowing the rate of context resulted in many fewer reports of the function words in a transcription task. Critically, blended function words in acoustically identical portions of the speech stream elicited an anterior and central first negative peak (N1) and N280 only in the condition that promoted reports of those words (fast casual context compared to slowed context). These results show that distal speech rate affects early perceptual processing and suggest that speech rate influences listeners' state before they encounter the minimal acoustic cues associated with spoken but blended words. These results have implications for understanding the role of speech rate in processing a broad range of speech signals, including those that lack clear acoustic cues to word onsets.

Key words: Speech perception, word segmentation, distal speech rate, ERP, N100

1. Introduction

Understanding speech requires using spectral and temporal information in complex and variable signals to extract content. This process is not limited to categorizing proximal acoustic features into phonological categories. Instead, listeners incorporate prosodic cues at multiple timescales to determine phonemes, words, and boundaries between words in continuous speech (Baese-Berk et al., 2014; Cho, McQueen, & Cox, 2007; Christophe, Peperkamp, Pallier, Block, & Mehler, 2004; Miller & Liberman, 1979; Salverda et al., 2007). Strikingly, distal speech rate – that is, the rate of speech context which is nonadjacent to a to-be-detected word – has been shown to affect whether listeners report hearing entire words or not (Dilley & Pitt, 2010). As such, there is ample evidence that distal speech rate affects language processing (Brown, Salverda, Dilley, & Tanenhaus, 2015; Dilley, Morrill, & Banzina, 2013; Heffner, Dilley, McAuley, & Pitt, 2013; Lai & Dilley, 2016; Morrill, Baese-Berk, Heffner, & Dilley, 2015). However, the current understanding of the mechanism by which distal speech rate affects language processing is insufficient. This work uses ERP measures to arbitrate between models of speech processing that posit a role for speech rate in early perceptual processing and those that locate speech rate effects in later stages of processing.

Of course, distal speech rate is not the only cue listeners use to determine word boundaries in continuous speech. Word onsets are also syllable onsets, and the characteristic arrangement of consonants and vowels into syllables has statistically reliable consequences for acoustic properties of speech. Consonants – which form onsets and codas of syllables – typically have substantially lower amplitude than vowels (Stevens, 2002). Hence, speech shows alternation between higher-amplitude syllable nuclei and lower-amplitude syllable edges at a frequency of approximately 4-8 Hz (Elliott & Theunissen, 2009; Giraud & Poeppel, 2012; Peelle

& Davis, 2012), referred to as the amplitude envelope. Amplitude envelopes have been shown to be important for extracting lexical content, segmental information, and speech rate from speech signals (Drullman, 1995; Drullman, Festen, & Plomp, 1994; Fogerty, 2013; Fogerty & Humes, 2012; Ghitza & Greenberg, 2009; Peelle & Davis, 2012). Coordinated neuronal activity can be observed to oscillate at frequencies associated with the speech amplitude envelope; such oscillations are thought to facilitate speech decoding (Ding, Melloni, Zhang, Tian, & Poeppel, 2016; Giraud & Poeppel, 2012; Zion Golumbic, Poeppel, & Schroeder, 2012).

In addition to those marked by changes in amplitude, word onsets with minimal amplitude dips or other spectrotemporal discontinuities occur frequently in natural speech. These acoustically weak onsets provide a medium for observing the effects of distal speech rate on language processing. For example, words like *are, our*, *a, an,* and *or* are some of the most common in spoken language. However, vowel-initial words, and especially vowel-initial function words, frequently lack acoustic discontinuities with preceding speech (Dilley, Shattuck-Hufnagel, & Ostendorf, 1996; Pierrehumbert & Talkin, 1992; Redi & Shattuck-Hufnagel, 2001). Semivowel consonants, like /r/, /l/, and /w/, are highly sonorous and have vowel-like character, evidencing a more continuous profile of acoustic change with respect to surrounding vocalic segments (Stevens, 2002). Further, in conversational speech, consonants may be elided or lenited (Johnson, 2004; Lavoie, 2002; Shockey, 2008), leading to the reduction or even elimination of local acoustic discontinuities (Baese-Berk, Dilley, Schmidt, Morrill, & Pitt, 2016; Dilley, Arjmandi, & Ireland, 2017; Dilley, Arjmandi, Ireland, Heffner, & Pitt, 2016). Coarticulation can be so extreme that the spectra of adjacent syllables blend partially or fully (Dilley et al., 2017; Dilley et al., 2016; Dilley & Pitt, 2010; Johnson, 2004; Shockey, 2008).

In seminal work, Dilley and Pitt (Dilley & Pitt, 2010) established that speech rate indeed plays an important role in recognizing acoustically weak word onsets. Spoken sentences included coarticulated, reduced function words (e.g., *or* spoken as “errr” in *Deena doesn’t have any leisure or time*). When presented in the fast casual context in which they were recorded, participants’ transcriptions of the speech contained the critical words on a majority (79%) of trials. However, when the distal context (e.g., *Deena doesn’t have any lei … ime*) was slowed while holding constant the proximal portion (i.e., the critical function-word containing region and adjacent segments, e.g., *…sure or t…*), participants reported many fewer (33%) critical function words. Effectively, words with acoustically weak onsets disappeared as a result of slowing the context speech rate (Dilley & Pitt, 2010). Moreover, the complementary effect was found for sentences that were spoken without the critical function word (e.g., *Deena doesn’t have any leisure time*). When the utterances spoken without function words were presented in the context in which they were recorded, participants’ transcriptions seldom contained the critical words. However, speeding the distal context while holding constant the proximal portion (e.g., *…sure t…*) resulted in listeners’ transcribing seven times more critical function words.

Subsequent studies showed that distal speech rate has a remarkably powerful effect on what listeners report hearing, even “overcoming” proximal acoustic discontinuities and other conflicting information about word boundaries (Baese-Berk et al., 2014; Dilley et al., 2013; Heffner et al., 2013; Heffner, Newman, Dilley, & Idsardi, 2015; Lai & Dilley, 2016; Morrill et al., 2015; Morrill, Dilley, McAuley, & Pitt, 2014). Starting with phrases spoken with highly reduced, coarticulated words, Heffner et al. (Heffner et al., 2013) manipulated distal speech rate by slowing it to varying degrees while parametrically crossing different degrees of manipulated acoustic discontinuity (in four experiments involving amplitude decrease, consonant noise, word duration, and fundamental frequency decrease) at the onset of the coarticulated words. Notably, even with substantial proximal acoustic discontinuity at critical word onsets, distal speech rate had a strong influence on whether or not the words were included in transcriptions. For example, even with the onsets of critical words marked by a significant amplitude drop of 18 dB relative to the original stimulus, slowing distal speech rate resulted in up to a 40% drop in reports of those critical words (Experiment 1, Heffner et al., 2013).

Combining the established paradigm for detecting effects of distal speech rate on words reported in transcriptions with measurements of ERPs time-locked to the onset of acoustically weak function words will permit a determination of whether speech rate plays a perceptual or post-perceptual role. For example, representations of speech rate could result in predictions about when upcoming word onsets are likely to occur, putting the listener in a different state before encountering the acoustically weak onsets. Listeners' representations of the probability that a word onset is to occur could then affect perceptual processing of incoming acoustic information. Such effects on perceptual processing would be indexed by ERPs in the first 200 ms after onset. Auditory evoked potentials in this time window, including the characteristic P1-N1-P2 oscillations in response to acoustic onsets, are modulated by both physical characteristics of the stimuli and the state listeners are in when a sound occurs (Hillyard, 1981; Hillyard, Hink, Schwent, & Picton, 1973; Lange, Rösler, & Röder, 2003; Sanders & Astheimer, 2008; Woods & Alain, 2001). In contrast, listeners may hold multiple possible interpretations of the acoustic information they have already encountered, and use contextual information to arbitrate between those possible interpretations. In that case, the influence of speech rate information on which interpretation processing ultimately selects would not be expected to be tightly time-locked to acoustic information, and therefore would not be evident in ERPs.

Two lines of evidence make it plausible that distal speech rate might affect early perceptual processing of acoustically weak function word onsets. First, it has been shown that word onsets in natural, continuous speech tend to elicit larger auditory evoked potentials than acoustically similar syllable onsets (Sanders & Neville, 2003). The same is true of physically identical syllables and sounds in artificial languages; sequence onsets elicit larger auditory evoked potentials only after listeners acquire the information needed to segment continuous streams (Abla, Katahira, & Okanoya, 2008; Sanders, Ameral, & Sayles, 2009; Sanders, Newport, & Neville, 2002). These studies demonstrate that ERPs are sensitive to online speech segmentation driven by cues other than local changes in amplitude envelopes. Second, a different kind of distal prosodic cue based on pitch and timing affects how listeners segment subsequent lexically ambiguous syllable sequences (e.g., *crisis turnip* vs. *cry sister nip*). Acoustically identical syllable onsets elicit larger P1-to-N1 deflections when they follow the prosodic context that supports their interpretation as word onsets (e.g., *nip* rather than *turnip*) (Breen, Dilley, McAuley, & Sanders, 2014). If distal speech rate affects perceptual processing, it will also affect ERPs within 200 ms after onset. Acoustically weak syllable onsets that are reported to be word onsets are predicted to elicit an auditory onset response in ERPs; those same weak onsets that are reported to be continuations of words due to slowing of the context speech rate are predicted to elicit little or no response.

ERPs also index many post-perceptual processes. For example, closed-class function words and open-class content words elicit distinct ERPs at times more than 200 ms after onset (Neville, Mills, & Lawson, 1992). Function words elicit an N280 whereas content words elicit an N400. ERPs time locked to acoustically weak function word onsets might reveal an N280 under conditions that promote reports of those function words (fast, casual context). However, the ability to observe any differences in ERPs based on context speech rate is dependent on post-perceptual processes being precisely time locked to the availability of acoustic information in the speech stream. If the context speech rate effects observed in transcriptions reflect an influence on selection among multiple possible interpretations, variability in the timing with which the system reaches resolution might be great enough to preclude differences in ERPs.

In the present study, listeners were asked to transcribe phrases that were semantically and syntactically well-formed with or without a critical function word. A listener heard each phrase a single time, either at the fast casual rate at which it was recorded or with the distal context slowed but the critical function word and adjacent segments left untouched. To the extent that context speech rate affects online perceptual processing that is time locked to acoustically weak word onsets in continuous speech, ERPs elicited by physically identical signals will differ depending on the speech rate context in which they are presented. Specifically, function words with acoustically weak onsets presented in the context that encourages their inclusion in transcriptions (fast, casual context) will elicit larger auditory evoked potentials (N1) and a component associated with closed-class words (N280). In contrast, when the context speech rate is slowed causing the same acoustic information to be interpreted as part of the final syllable of the previous word, ERP indices of onsets and function words will be absent. Alternatively, if context speech rate affects only later processing, but with consistent timing relative to the ambiguity in the speech stream, differences in ERPs will appear only later in the time series. Finally, if there is a great deal of variability in the timing of resolution of multiple possible interpretations of the speech signals, there may be no differences in ERPs time locked to acoustically weak function word onsets. Thus, the results are expected to provide important information about the mechanisms by which distal speech rate affects speech processing.

2. Method

2.1 Participants

Eighteen adults (10 women and 8 men, mean age = 22 years) contributed data. Another nine adults participated in the experiment but did not provide artifact-free data from a sufficient number of trials in both conditions (≥ 30). All participants were right-handed, native-English speakers, with no known hearing or neurological impairment. They provided informed consent and were compensated for their time at a rate of $10 per hour.

2.2 Stimuli

Stimuli in the present experiment were constructed from the corpus of recordings elicited as part of Experiment 1 in Dilley and Pitt (Dilley & Pitt, 2010). On each trial in that experiment, subjects were shown a sentence prompt on a computer screen that subsequently disappeared, and they then had to produce the sentence from memory. This task reliably elicited scripted sentences that were very frequently produced in a casual style. Crucially, the sentences were constructed to contain critical function words that could be spoken with a vowel-initial reduced variant (e.g., *her* or *or spoken as* /ɚ/) positioned after a word ending in an open syllable or rhotic diphthong.

This combination elicited frequent productions of weak onsets for the critical function words with near-total blending across the word boundary. Sentences were grammatically felicitous whether or not the critical function word was present for at least one word following the function word. 79 tokens of sentences were selected from the corpus; these materials were spoken by 22 different talkers (*M* = 3.6 sentences per talker). Previous studies drawing on the same corpus have shown that experimental items containing critical function words with weak onsets are reported in 79%-83% of transcriptions when presented at the fast, casual speech rate at which the items were recorded (Baese-Berk et al., 2014; Dilley & Pitt, 2010). The sentences selected for the current experiment are given in the Appendix. For each of the sentences, any ending word(s) which disambiguated the grammatical structure were truncated. The result of this step served as the basis for the subsequent manipulations described below. Fifty fillers from the same corpus that were not constrained to have function words in particular phonetic contexts were also selected.

Each target region was defined as the critical function word, plus the preceding syllable and the following phoneme. Distal context was defined as the remainder of the speech not including the target region. The time points marking the start and end of the target region were annotated in Praat (Boersma & Weenink, 2017) by referencing spectrotemporal characteristics in spectrogram and waveform displays. We used phonetic segmentation criteria from the Buckeye corpus (Pitt et al., 2007) and also adhered to standard phonological assumptions about resyllabification (e.g., the maximal onset principle) (Kahn, 1976; Selkirk, 1981). Next, PSOLA resynthesis was applied to each of the 79 phrases to create two distal context conditions as shown in Figure 1. For the *normal context* condition, a multiplicative factor of 1.0 was applied to the duration of the entire fragment. For the *slowed context* condition, a piecewise-linear function was constructed that indicated the time-varying multiplicative factor. The factor was 1.0 during the target region, as it was for the normal context condition, and 1.9 for most of the distal context. The function included a 10 ms ramp from 1.9 to 1.0 prior to the target region and a 10 ms ramp from 1.0 to 1.9 immediately following the region. For the normal context condition, the target region began 857 ms (range 371- 1543 ms) after context onset. As a result of the speech rate manipulation, for the slowed context condition the critical region began 1605 ms (range 693 - 2932 ms) after context onset. The reduced function words began 41-342 ms (M = 137 ms, SD = 54 ms) after the start of the target regions. The large variability in length of time between the end of the preceding context and onset of the function word ensured that ERPs time-locked to the change in speech rate in the slowed context phrases could be separated from ERPs time-locked to the onset of the reduced function words.

Times at which the stimuli transitioned from context to the target region (when speech rate changed in the slowed context condition) and onset times of the reduced functions words were initially measured by three coders who were unaware of the experimental manipulation. Coders were asked to view the spectrographs and listen to the phrases using a gating procedure to determine "the earliest indication of the specified syllable." For items on which the times provided by the three coders had a range less than 12 ms (N = 76 target region onsets, N = 24 function word onsets), ERPs were time-locked to the earliest of the coders' times. For the remaining items, one of the authors (LDS) followed the same procedure to provide an additional measure of onset time. If this new measurement fell within a 12 ms range with times provided by at least two of the coders, ERPs were time-locked to the earliest of these three times (N = 81 target region onsets, N = 97 function word onsets). For the remaining items, LDS provided a new measurement while blind to her previous numbers. The two measurements provided by LDS fell within 12 ms of each other and the time provided by at least one of the coders for all of these items (N = 1 target region onset, N = 37 function word onsets); ERPs were time-locked to the earliest of these times.

2.3 Procedure

Participants were asked to listen to the speech fragments and to type what they heard at the end of each trial. Two lists were created such that each participant heard half of the items with the normal speech rate context, half with the slowed context, and all 50 filler phrases. The items that were presented in the normal context condition for half of participants were presented in the slowed context condition for the other half. The phrases were presented in a different random order for each participant. A fixation cross was shown 120 cm in front of participants; they were instructed to look at this symbol and avoid blinking during presentation of the phrases. After a phrase was complete, a prompt to type what was heard was presented on the monitor. Participants were able to see what they had typed and make any corrections until they hit the 'complete' button. However, phrases were never repeated and transcriptions could not be changed after indicating they were complete. Participants were encouraged to take three short breaks during the experiment.

Electroencephalogram (EEG) was recorded with 128-channel HydroCel Geodesic Sensor Nets (Electrical Geodesics, Inc., Eugene, OR) using a bandwidth of 0.01–100 Hz and a sampling rate of 250 Hz. Electrode impedance was reduced below 50 kΩ at the beginning of the experiment and was maintained below 100 kΩ throughout. A 60 Hz filter was applied before segmenting the EEG into 500 ms epochs. Each epoch began 100 ms before the onset of the phrase, the target region, or the function word. Epochs that contained artifacts from blinks, eye-movements, or skin potentials were excluded from analysis. Data from each participant recorded at each electrode were averaged separately by context condition. ERPs were then re-referenced to the averaged mastoid measurements and baseline corrected to the mean amplitude during the initial 100 ms.

2.4 Analysis

Behavioral data were analyzed for all 18 participants who provided sufficient ERP data. Transcriptions that included all of the words in a phrase with or without the critical function word were considered to have been completed. Words were considered correct even if they were misspelled. There were no differences in the percentage of correct, completed transcriptions for the two context conditions (normal = 94.6%, slowed = 93.9%), indicating that the context manipulation neither promoted nor hindered comprehension. The percentage of transcriptions that included the critical function words was calculated based on completed responses only.

To test the hypothesis that context speech rate affects early perceptual processing of subsequent speech sounds, mean amplitude measurements were taken 100-150 ms (N1) and 200-300 ms (N280) after the onset of the function words. To determine if any observed differences between conditions could be explained by ERPs elicited before onset of the function words, mean amplitude measures were also taken on waveforms time-locked to the onset of the phrases (80-130 ms and 180-250 ms) and to the onset of the target regions when speech rate shifted from slow to normal in the slowed-context condition (0-150 ms and 150-400 ms). Measurements from 81 electrodes were grouped into nine scalp regions of nine electrodes each and included as two electrode position factors in analysis: Anterior/Central/Posterior (ACP) and Left/Medial/Right (LMR). The specific groups of electrodes included in each level of the electrode position factors are shown in Figure 2. A repeated-measures ANOVA (Geisser-Greenhouse corrected) was applied to each mean amplitude measurement: Context Speech Rate (normal, slowed) x ACP x LMR.

3. Results

3.1 *Behavioral findings from Transcription Task*

The percentage of responses that included the critical function word was computed based on correct, completed transcriptions. When listening to phrases at the normal speech rate at which they were recorded, the majority (M = 77.1%, SD = 13.5%) of transcriptions included a critical function word. By contrast, when the distal context speech rate was slowed, the same physical target region resulted in much less frequent reports of a critical function word (M = 13.5%, SD = 10.3%). Further, this distal rate effect was consistent across individuals; every participant reported hearing the reduced function word on a much larger proportion of the normal speech rate trials than of the slowed context trials.

3.2 *Event-Related Potentials*

As shown in Figure 3, distal speech rate affected perceptual processing of critical function words contained within acoustically identical chunks of speech held constant across context conditions (the target region). First, function words elicited a first negative peak (N1) 100-150 ms after onset only in the normal rate condition that encouraged the inclusion of these words in transcriptions ($F(1,17) = 16.7$, $p < .001$). This N1 effect was larger over anterior and central regions (Context Speech Rate x ACP x LMR: $F(4,68) = 3.5$, $p = .01$; Context Speech Rate at anterior and central only, $F(1,17) = 18.2$, $p < .001$). In the next time window (N280) 200-300 ms after critical function word onset, the mean amplitude measure was also more negative in the normal compared to the slowed context condition ($F(1,17) = 12.7$, $p < .005$). This effect was also largest over anterior and central regions (Context Speech Rate x ACP: $F(2, 34) = 4.1$, $p < .01$; Context Speech Rate x ACP x LMR: $F(4, 68) = 4.0$, $p < .01$; Context Speech Rate at anterior and central only, $F(1,17) = 22.3$, $p < .0001$).

ERPs time locked to phrase onsets (Figure 4) were examined to confirm the scalp distribution of auditory onset components and to detect any differences in waveforms elicited by normal and slowed acoustic onsets. As is typical of any acoustic onset, phrase onsets elicited a series of positive-negative-positive peaks. The first negative peak (N1, 80-130 ms) was largest over anterior and central regions. Moreover, onsets of normal rate phrases elicited a larger N1 than onsets of slowed rate phrases ($F(1,17) = 10.7$, $p < .01$).

ERPs elicited by the onsets of physically identical target regions were also expected to differ by context since speech rate changed in the slowed context condition (multiplicative factor of 1.9 to 1.0 across 10 ms) but not in the normal context condition (multiplicative factor of 1.0 for context and target region). Speech rate transitions in the slowed context condition elicited a positivity 0-150 ms after onset ($F(1,17) = 12.8$, $p < .01$) that continued into the later measurement window 150-400 ms after onset ($F(1,17) = 26.1$, $p < .001$). The later positivity was larger over anterior and central regions (Context Speech Rate x ACP: $F(2, 34) = 8.0$, $p < .001$; Context Speech Rate x ACP x LMR: $F(4, 68) = 3.7$, $p < .01$; Context Speech Rate at anterior and central only, $F(1,17) = 33.0$, $p < .001$).

4. Discussion

The present results are the first evidence that distal context speech rate affects online neural responses. Importantly, different ERP waveforms were elicited by acoustically identical portions of natural speech signals as a function of changes in context speech rate. These data are the first definitive evidence that distal speech rate affects early perceptual processing. Acoustically weak onsets elicited an N1 and N280 *only* when the context speech rate supported perception of the acoustic information as entire function words rather than ends of previous words. The effect of distal speech rate on perceptual processing is potentially sufficient to explain the large effects on whether or not listeners include the critical function words in transcriptions. Further, the results shed light on the problem of how listeners recognize the onsets of syllables and words in the absence of local amplitude changes and/or other spectrotemporal discontinuities. Distal speech rate determines whether or not weak onsets in continuous speech elicit ERPs associated with acoustic onsets (i.e., the auditory N1), an effect that is equivalent to introducing a small amplitude change. Indeed, distal speech rate may be the only cue that allows listeners to recognize acoustically blended onsets during early perceptual processing.

These results clearly distinguish between possible explanations of effects of distal speech rate on whether or not listeners report hearing reduced function words in transcriptions; distal speech rate affects early perceptual processing. Further, differences in neural responses to acoustically weak function words by 100 ms after onset suggest that listeners were in a different state in the normal and slowed speech rate conditions *before* encountering the acoustically ambiguous onsets. The N1 effects in the current study are consistent with previous data showing that speech segments which are perceived as word onsets elicit larger auditory evoked potentials than acoustically similar segments which are perceived as word-internal syllable onsets (Breen et al, 2014; Sanders & Neville, 2003; Sanders et al., 2002). Further, the early differences in perceptual processing of segments that are and are not word onsets are not dependent on the type of information that predicts upcoming onsets, as long as the information is made available before the onsets themselves.

We suggest this pattern of results may reflect predictive coding of the ongoing speech stream. Under predictive coding accounts of speech processing, beliefs about the likely lexical content of speech and the communicative goals of the speaker give rise to multiple, concurrent hypotheses about what caused specific sensory input, informed by statistical priors and modeled by hierarchical Bayesian inference (Friston, 2005; Rao & Ballard, 1999; Tavano & Scharinger, 2015). Applied to the current data, this framework would suggest that distal speech rate determines expectations about the timing of upcoming word onsets. Speech is proposed to be chunked into sonorant units which are delimited by robust spectrotemporal discontinuities (Stevens, 2002). The distribution of "chunk" durations in combination with information about the number of linguistically meaningful units in each chunk could be used to predict the probability that incoming acoustic information is the onset of a new syllable or word. In the current study, when listeners heard a fast, casual context and the acoustically marked onset of the target region, they would be expecting an onset around the time of the critical function word. In contrast, when listeners heard slowed context with larger chunk durations, they would not expect a new onset until farther into the target region and after the critical function word.

The effects of expecting segments in continuous speech to be word onsets on early auditory evoked potentials are remarkably similar in timing, amplitude, and distribution to ERP effects of auditory selective attention (Hillyard, 1981; Hillyard et al., 1973; Lange et al., 2003; Sanders & Astheimer, 2008; Woods & Alain, 2001). In fact, listeners have been shown to allocate greater attentional resources to extraneous sounds presented at the same time as word onsets in continuous speech (Astheimer & Sanders, 2009). These findings raise the possibility that in the current study, listeners were allocating more attention to the time of the ambiguous function word onset following the normal speech context than following the slowed speech context. Dynamic Attending Theory (DAT) might explain how distal context speech rate affects how much attention was directed to the onsets of critical function words. DAT postulates that the amount of attention directed to a stimulus at any point in time is influenced by external periodicities (Jones, 1976; Keele, Nicoletti, Ivry, & Pokorny, 1989; Large & Jones, 1999; McAuley & Jones, 2003; Pashler, 2001; Schulze, 1978). Recent evidence shows that rhythms in non-linguistic stimuli can direct selective attention in a manner that affects early perceptual processing (Fitzroy & Sanders, 2015). However, for distal speech rate to reliably influence reports of words with acoustically weak onsets, the context must be intelligible (Pitt, Szostak, & Dilley, 2016). DAT cannot explain why distal speech rate effects on behavioral responses require the higher-level linguistic information available in intelligible speech and absent in tones, low-pass filtered speech, noise-vocoded speech, and sine-wave speech.

Finally, the second negative peak 200-300 ms after the onset of ambiguous function words following the normal context, which more often resulted in inclusion of those words in transcriptions, is likely related to the N280 reported in response to closed-class words (Neville et al., 1992). It is not at all surprising that the neurophysiological marker of processing closed-class words would be evident when listeners report hearing those words compared to when they do not. However, the fact that it is observed with the same early timing reported for words presented in isolation suggests that the acoustic ambiguity of these function words does not slow or significantly alter processing. Instead, the effects of context speech rate on processing of the ambiguous function words appears to be equivalent to listeners hearing unambiguous evidence of closed-class words. Listeners are not deciding that such a word was plausibly spoken when they are constructing their transcriptions. Rather, context speech rate plays a role in early perceptual processing such that listeners do not consider the presence or absence of the function words to be at all ambiguous.

The finding that distal speech rate affects early perceptual processing of acoustically weak function word onsets has several important implications for speech processing models. First, it suggests that processes to extract rate information from speech signals are in use under typical listening conditions. Other research suggests that amplitude envelopes from intelligible speech play an important role in listeners' representations of speech rate information (Ding et al., 2016; Drullman et al., 1994; Elliott & Theunissen, 2009; Giraud & Poeppel, 2012; Pitt et al., 2016; Zion Golumbic et al., 2012). Second, the results indicate that predicting the timing of upcoming, linguistically relevant onsets is ongoing during speech processing. We suggest these predictions influence the temporal dynamics of attention allocated to speech, as well as the interpretation of incoming acoustic information. Third, the pattern of results across this and other research (Breen et al., 2014; Sanders & Neville, 2003; Sanders et al., 2002) indicate that factors which influence predictive processes, including speech rate, can be functionally equivalent to acoustic modulations. In the current study, differences in ERPs driven by distal speech rate were identical to what would be predicted for making function words more and less acoustically salient (N1 and N280). Additionally, the results speak to the under-studied problem of how listeners perceptually recover words with minimal acoustic discontinuity to cue their onset. The use of speech rate information to predict the timing of upcoming onsets is a necessary part of successful language comprehension when acoustic cues to onsets are lacking.

Acknowledgements

This research was supported by the National Science Foundation under grant BCS-1431063 to L. Dilley, BCS-1431118 to L. Sanders, and BCS-1559719 to N. Viswanathan. We are grateful to Matt Lehet for helpful feedback on an earlier draft of the paper. We thank Andrea Hulme for assistance with stimulus preparation, Daniela Kucher and Evangelos Souliotis for data collection, and Madeleine McAuley and Leif McAuley for general project support.

Figure Captions

Figure 1. Example of a speech stimulus. Critical regions were not altered from the original fast, casual speech when context was slowed.

Figure 2. Approximate positions of scalp electrodes. Data from nine nearby electrodes were averaged together before mean amplitude measurements were made. Scalp position was included in omnibus ANOVAs as two three-level factors.

Figure 3. ERPs time locked to function word onsets. Acoustically weak function word onsets elicited an N1 and N280 only when presented in the normal-rate context that encouraged inclusion of those function words in transcriptions.

Figure 4. ERPs time locked to phrase onsets. Abrupt acoustic onsets at the beginning of phrases elicited a typical auditory onset response (P1-N1-P2). N1s were larger in response to the onsets of contexts presented at the normal (faster) speech rate.

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Appendix: Experimental stimuli used in the present experiment. The portion between vertical lines indicates the target region of acoustic material that was held constant across the normal and slowed context speech rate conditions. Words shown in parentheses indicate the plausible completion of sentences that were included in initial recordings but truncated during stimulus creation due to the fact that they grammatically disambiguated the function word as being present. Resulting sentence fragments were grammatically compatible with either the critical function word being present or absent.

1. The Smiths wouldn't |buy a B|utterball (turkey.)
2. It's not easy to con|vey a l|ikely (position.)
3. Jan would de|lay a l|imited (amount of time.)
4. Bill didn't want to |fly a l|onger (distance.)
5. The boy wanted to |glue a b|roken (toy.)
6. It makes no sense to o|bey a p|etty (law.)
7. John said he would o|bey a r|ebel (leader.)
8. Mark said he would pur|sue a l|iberal (vote.)
9. Dave asked how long it takes to re|pay a l|arge (debt.)
10. It takes a lot of work to re|view a p|ersonal (file.)
11. Anne wanted to |see a v|ery funny (movie.)
12. It costs a lot to tat|too a p|ink (flamingo.)
13. Dawn said that it's easy to go |to a r|egular (store.)
14. Sally might |try a l|iquid (detergent.)

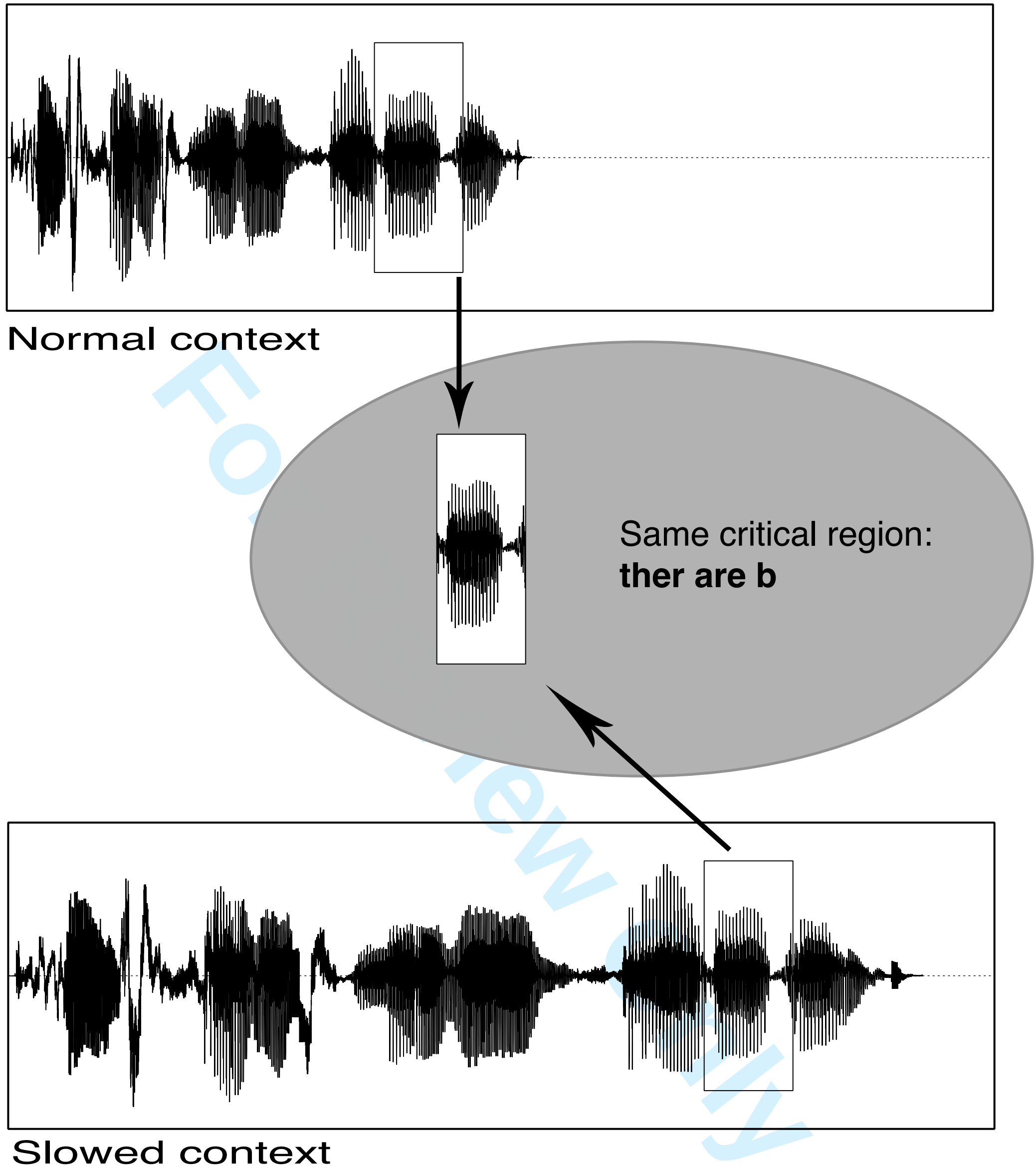
15. Frank thinks that sadness and an|ger are b|oth (bad.)
16. George thought my father and broth|er are l|ike good (friends.)
17. Ruth saw the maid and but|ler are a|t the top (of the stairs.)
18. Connor knew that bread and but|ter are b|oth (in the pantry.)
19. Chris said his mother and fa|ther are b|oth (old.)
20. Glenn thought his friend and neigh|bor are l|ike plenty (of others.)
21. Taylor knew the principal and teach|er are f|rom Ohio (.)
22. Sam knew |there are a|pples (which are sweet.)
23. Sam saw |there are c|lothes (which they liked.)
24. Rose knew that |there are l|amps (which are expensive.)
25. Tess thought that |there are l|oans (that have better rates.)
26. Sue said |there are l|unches (that are healthy.)
27. Todd said |there are r|ooms (which are ugly.)
28. Bob said |there are s|tocks (to buy.)
29. Zach knew that |there are th|ings (in the closet.)

30. The sign was replaced af|ter her b|lack (car got stolen.)
31. The message was clear af|ter her b|lank (stare said it all.)
32. The leaves fell af|ter her g|reen (lawn dried up.)

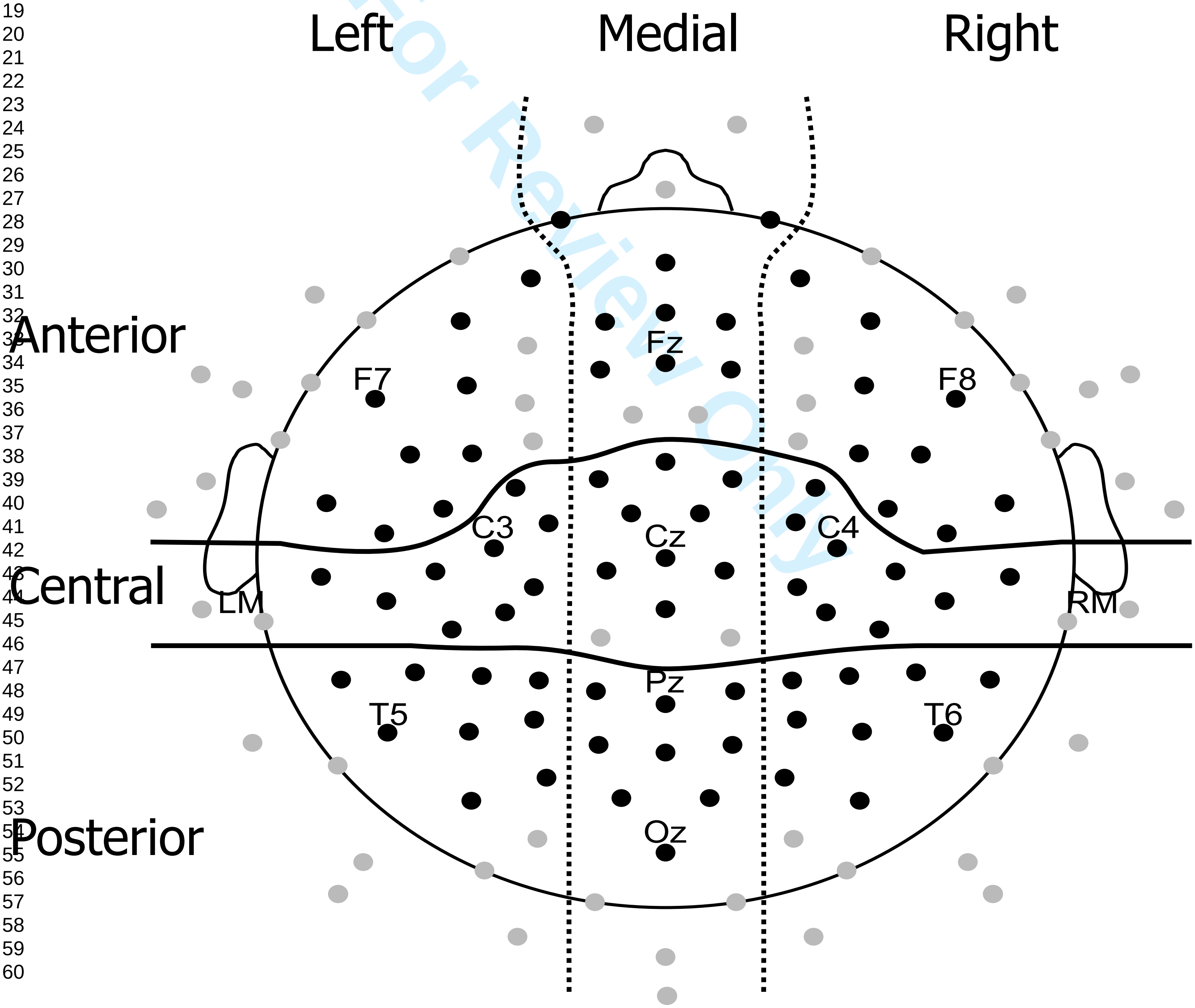
33. They were sad af|ter her p|oor (dog was put down.)
34. The value went up af|ter her r|ich neighbors (moved in.)
35. People were offended af|ter her r|ude (comments.)
36. Chris was very quiet af|ter her sh|arp (mind got in gear.)
37. The Perrys thought carefully af|ter her w|ise advice (was given.)
38. Dan took off af|ter her y|oung (friend was hired.)
39. It's not long be|fore her b|ad (back goes out.)
40. That should be done be|fore her g|ray hair (sets in.)
41. Lance said goodbye be|fore her l|arge (car got towed.)
42. Sam got it in writing be|fore her l|ast wishes (were granted.)
43. Lisa was done be|fore her l|oose curls (got tangled.)
44. It's not long be|fore her r|are wit (comes out.)
45. The manager hid the candy be|fore her s|ix kids (left.)

46. Steve pitched the ball to cen|ter or l|eft (last week.)
47. Marty gave him a dol|lar or t|wenty last week (.)
48. They promised him the fu|ture or a|id (last week.)
49. Don must see the har|bor or b|oats (this year.)
50. Jen's favorite line was "Hon|or or l|ife" most days (.)
51. John didn't tell the jun|ior or r|epresentative about it (.)
52. Deena doesn't have any lei|sure or t|ime (this week.)
53. Jack didn't vote for the mem|ber or c|onstituent (last week.)
54. Anyone must be a mi|nor or ch|ild (to enter.)
55. Dan couldn't watch the mon|ster or e|vil in the movie (.)
56. Tina decorated with pa|per or l|ace in most rooms (.)
57. George turned left at the riv|er or b|ank (last week.)
58. Sally sold all her sil|ver or j|ewelry last month (.)
59. Joan didn't want to face the stran|ger or a|ilment (this time.)
60. Jon called her su|gar or h|oney most of the time (.)
61. Fred would rather have a sum|mer or l|ake (house.)
62. We won't have any win|ter or w|et weather this year (.)

63. These houses |are our b|est (options for making money.)
64. Susan said those |are our b|lack socks (from the attic.)
65. Aspirin and other painkillers |are our d|rugs (of choice.)
66. These documents |are our f|ake (IDs to get into clubs.)
67. The Murrays |are our f|avorite (aunts and uncles.)
68. The callers |are our F|rench contacts (from the embassy.)
69. The Clines |are our g|ood friends (from college.)
70. Mom said these |are our g|ray gloves (from Aunt Bess.)
71. These copy machines|are our l|argest (ones.)
72. New laws |are our l|ast chances for a changing climate (.)
73. Those tickets |are our l|ate entries (for the sweepstakes.)
74. The Lees |are our o|ld friends from high school (.)
75. The snowsuits |are our o|nly (clothing.)
76. The Johnsons |are our r|ich relatives from California (.)
77. The accountants |are our w|ise advisors (on money matters.)
78. These birds |are our y|ellow canaries from New Zealand (.)
79. Phil and Mary |are our y|oung cousins (from Tennessee.)



Chris said his mother and fa**ther are b**oth (old.)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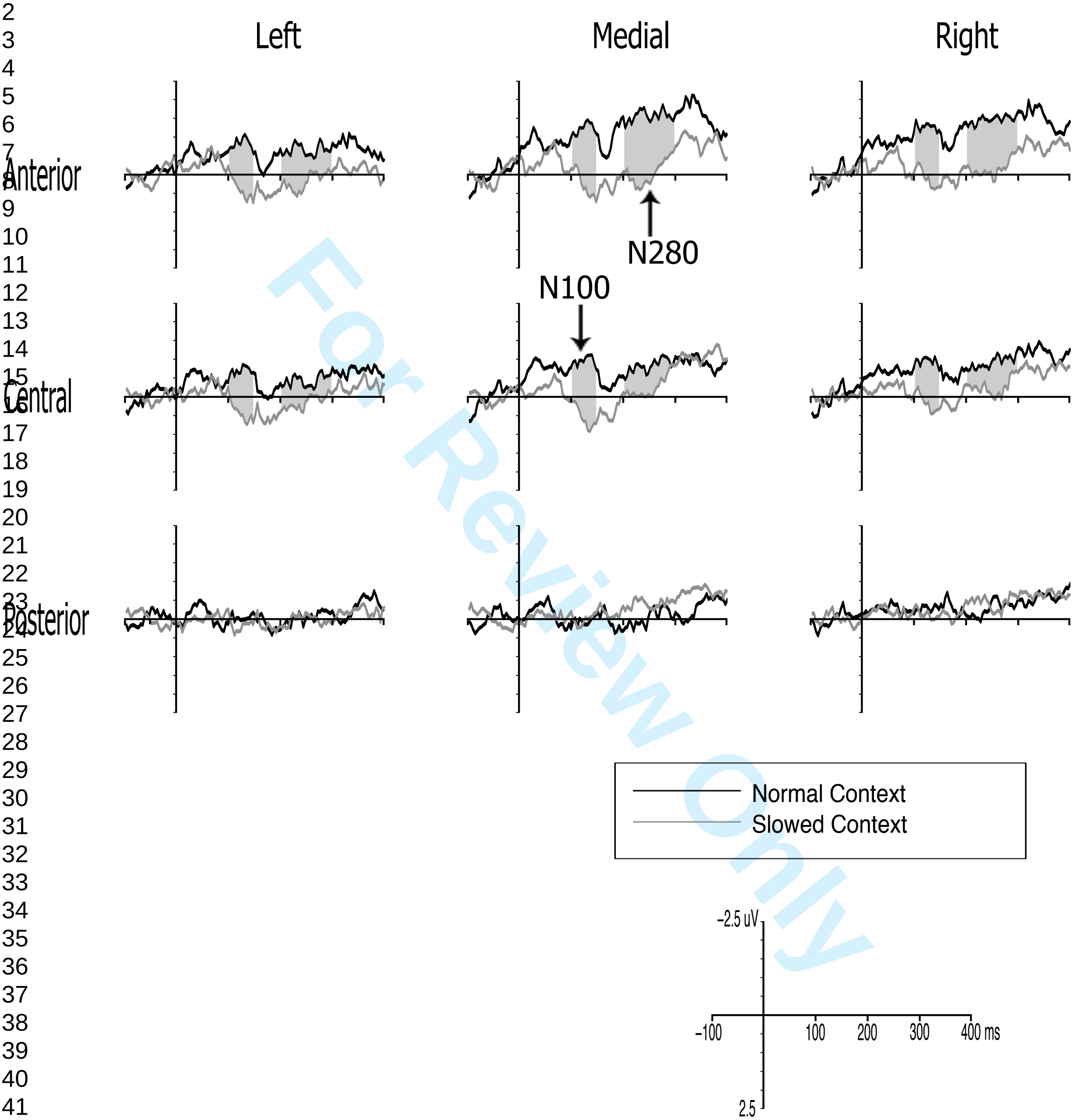
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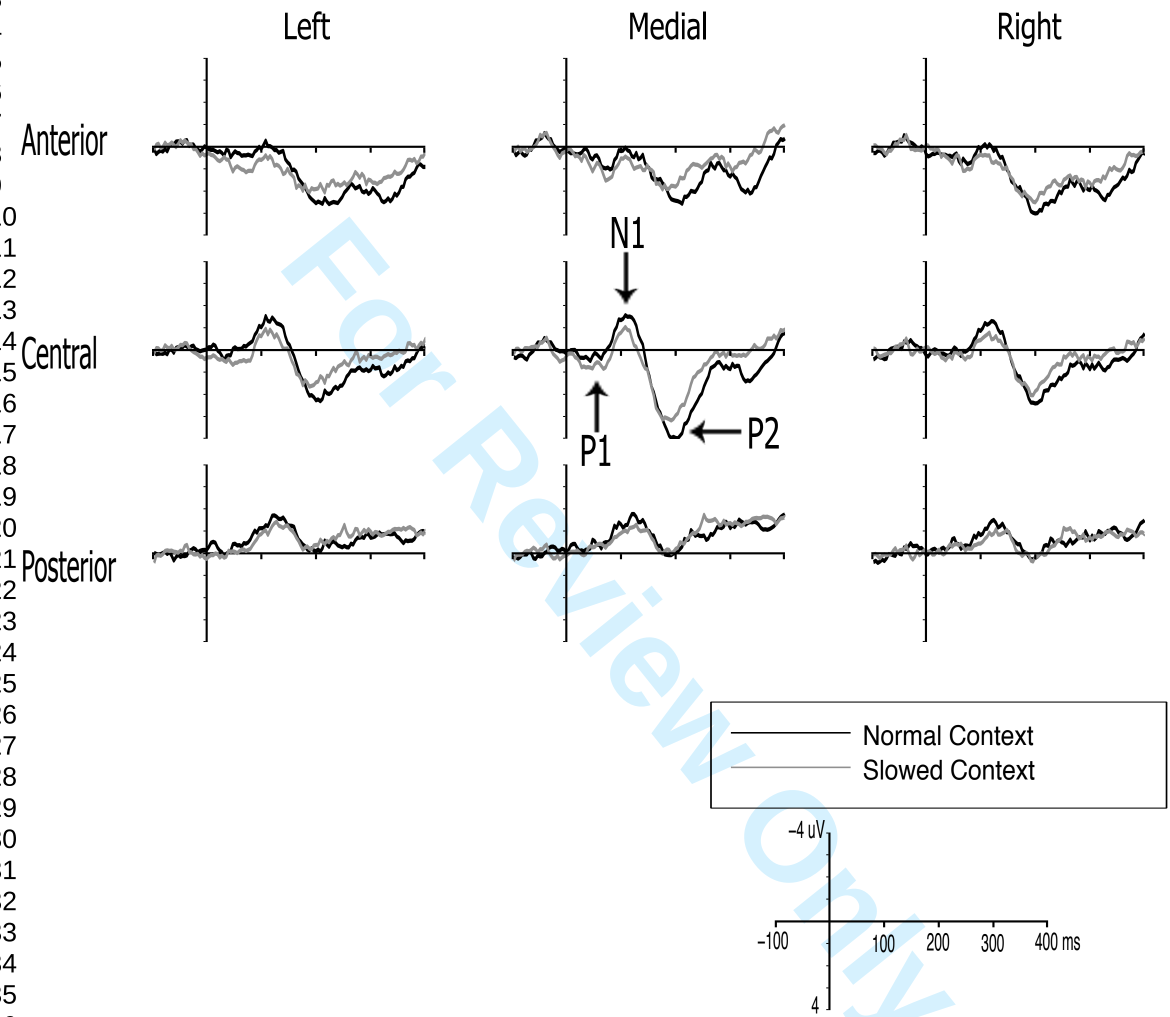
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